Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

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Introduction

- Customer churn refers to a customer leaving or stopping the use of a company's products or services.
- Customer churn can lead to reduced revenue and increased costs for acquiring new customers.
- Traditional approaches often depend on historical reports, manual analysis, and basic statistical methods.
- These approaches may not identify customers who are likely to leave early enough for preventive action.
- Our project uses Machine Learning to analyze customer information and predict whether a customer is likely to churn or stay.
- The proposed model is Elastic Net Regularized Logistic Regression, which combines L1 (Lasso) and L2 (Ridge) regularization.

Problem statement

Businesses have difficulty identifying customers who are likely to leave using traditional approaches such as:

- Historical reports
- Manual analysis
- Simple statistical methods

These approaches may not identify high-risk customers early enough to take preventive action.

Main Goal:

To develop a Machine Learning system that analyzes customer information and predicts whether a customer is likely to churn or stay, allowing businesses to identify potential churners at an early stage.

Objectives

- Collect relevant customer information for churn prediction.
- Understand and analyze the customer dataset.
- Clean and preprocess the collected data.
- Perform Exploratory Data Analysis (EDA) to identify patterns related to churn.
- Develop an Elastic Net Regularized Logistic Regression model.
- Evaluate the model using appropriate performance metrics.
- Provide a system that predicts Churn / No Churn and churn probability.

Dataset Overview

Dataset Used: Telco Customer Churn Dataset

- The dataset contains customer information from a telecommunications company.
- It includes details about customer demographics, services, contracts, and billing.
- The dataset contains 7,043 customer records and 33 columns.
- Our target variable is Churn Label, which tells us whether a customer has left the company or not.
- The target has two classes:
  - Yes → Customer churned
  - No → Customer stayed
- This is a Supervised Learning → Binary Classification problem.

Step 1 Load Dataset Import Libraries


```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Library	Purpose
Pandas	Load and manipulate the dataset
NumPy	Numerical calculations
Matplotlib	Create graphs and visualizations
Seaborn	Create statistical visualizations

- Loading the Dataset

```
from google.colab import files
upload = files.upload()
```

- Display the Dataset

```
df.head()
```

This displays the first 5 rows of the dataset.

- Check Dataset Size

```
print("Rows:",df.shape[0])
print("Columns:",df.shape[1])
```

Rows: 7043

Columns: 33

- Check Column Names

```
print(df.columns.tolist())
```

Step 2 Data Understanding

- Check Statistical Summary

```
df.describe()
```

This gives statistical information for numerical columns such as:

- Count
- Mean
- Standard deviation
- Minimum
- Maximum
- Quartiles

- Check Missing Values & Duplicate Values

```
df.isnull().sum()
```

```
df.duplicated().sum()
```

- Understand the Target Variable

```
df["Churn Label"].value_counts()
```

Churn Label	count
No	5174
Yes	1869

dtype: int64

Step 3 Data Cleaning

- Correct Total Charges Column

```
df["Total Charges"].dtype
df["Total Charges"] = pd.to_numeric(
```

In this dataset, Total Charges may be stored as an object/text instead of a numerical value.

- Remove Irrelevant Columns

```
columns_to_drop = [
    "Count",
    "Country",
    "State",
    "City",
    "Zip Code",
    "Lat Long",
    "Latitude",
    "Longitude",
    "Churn Value",
    "Churn Score",
    "Churn Reason"
]
```

These columns don't provide useful customer-behavior information for our prediction objective.

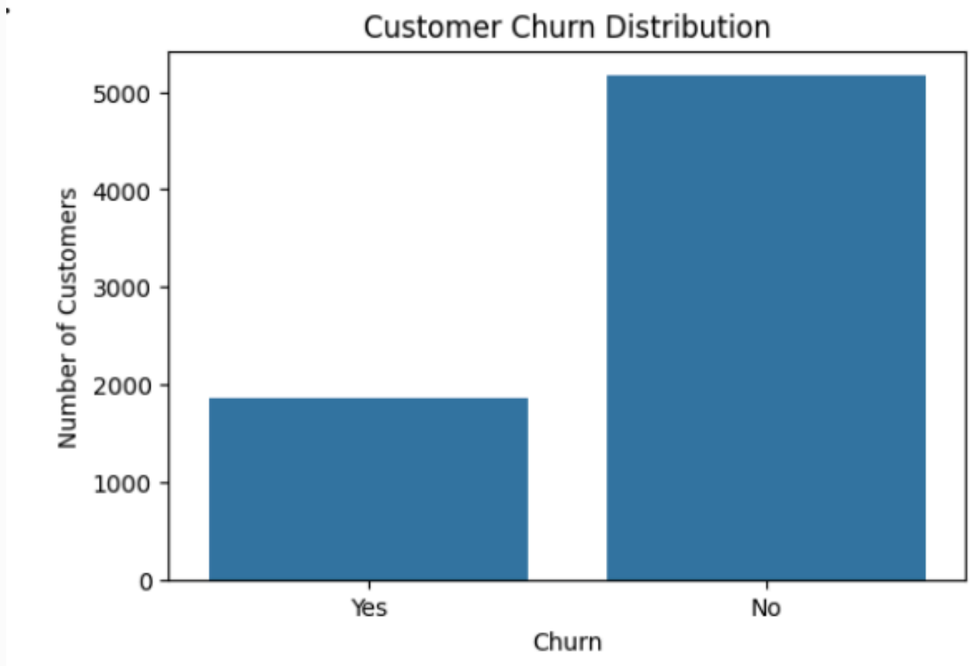
Step 4 EDA

- To understand the proportion of churned and non-churned customers.

```
plt.figure(figsize=(6, 4))

sns.countplot(data=df, x="Churn Label")

plt.title("Customer Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Number of Customers")
plt.show()
```



- Churn vs Contract Type

```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x="Contract", hue="Churn Label")

plt.title("Churn by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")
plt.xticks(rotation=15)
plt.show()
```

We analyze whether churn differs between:

- Month-to-month
- One year
- Two year

- Churn vs Tenure

```
#churn vs tenure
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Churn Label", y="Tenure Months")

plt.title("Churn vs Tenure")
plt.xlabel("Churn")
plt.ylabel("Tenure (Months)")
plt.show()
```

We compare the tenure distribution of:

- Churned customers
- Non-churned customers

- Correlation Heatmap

```
numeric_cols = df.select_dtypes(include=np.number).columns

numeric_cols
plt.figure(figsize=(10, 7))

correlation = df[numeric_cols].corr()

sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")

plt.title("Correlation Heatmap")
plt.show()
```

To understand the relationships between numerical variables.

THANK YOU